LRSC Single-Step Spectral Certificate v1.0

Prince Upadhyay - Independent Research

Locked baseline: M=99, k_idx=49, theta=0.08, A=0.50; 50 conjugate-balanced channels; relative complex L2 tolerance 0.001.

Result. The recorded fixed-cardinality global optimization places the first threshold crossing at **K_0.001(0.50)=11**. This is a computational statement for this locked one-step benchmark only.

Fresh direct reconstruction: N_act=81; full-channel closure error=2.3808015585084204e-16.

K	Recorded global subset	Direct error	Threshold
1	49	0.024191796924433	FAIL
2	48,49	0.021715391412619	FAIL
3	0,1,49	0.008528939402413	FAIL
4	0,1,48,49	0.002500311481442	FAIL
5	0,1,3,48,49	0.002413193978567	FAIL
6	0,1,46,47,48,49	0.001874507079545	FAIL
7	0,1,45,46,47,48,49	0.001735356145188	FAIL
8	0,1,44,45,46,47,48,49	0.001302595868239	FAIL
9	0,1,43,44,45,46,47,48,49	0.001189894735124	FAIL
10	0,1,42,43,44,45,46,47,48,49	0.001039820619099	FAIL
11	0,1,3,42,43,44,45,46,47,48,49	0.000962369127664	PASS

Threshold bracket. K=10: 0.001039820619099 > 0.001. K=11: 0.000962369127664 <= 0.001.

Global-optimality provenance. The subset identities are the recorded outputs of the completed global-search stage. This release independently rebuilds the operator and directly recomputes every listed residual. The bundled verifier is intentionally not misrepresented as a fresh exhaustive optimizer.

Structural observation. K=6 through K=10 contain channels {0,1} plus a contiguous high-index edge block expanding downward. At K=11, channel 3 enters: {0,1,3,42,43,44,45,46,47,48,49}. Its analytic origin remains open.

Claim boundary. No claim is made here of a physical law, universal/intrinsic dimension, emergent-spacetime validation, or multi-step dynamics. The certificate is tied to the locked operator, parameters, decomposition, norm, and tolerance.